American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$A\,(\,\overline{B\,C\;+\;\overline{\overline{A}}}\,)\;\;(\,\overline{\overline{\overline{A}}\,B\,C}\,)$$

A. 1
B. 3
C. 4
D. 5
E. 7

2. Data Structures

Find the internal path length of the binary search tree for:

OLDSMOBILECUTLASS

A. 41
B. 44
C. 46
D. 48
E. 54

3. Data Structures

There are 15 people who are in line for tickets to Taylor Swift's concert and there are 3 ticket booths. Customers are served in the order given at the first ticket booth available. The first 3 customers (Amy, Beth, and Cam) start at booths 1-3 in that order when the booths open.

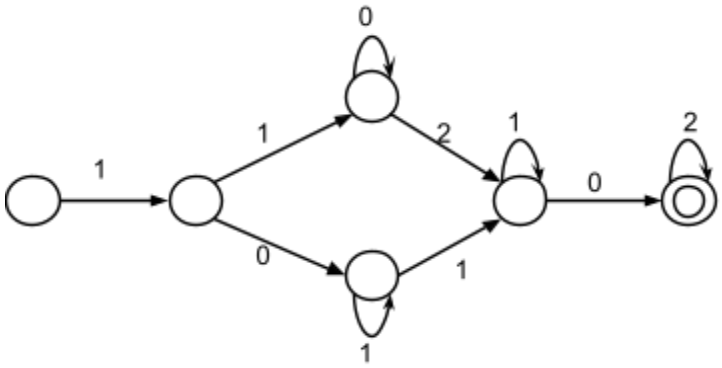
The times for each customer being at the ticket booth are as follows:
Amy = 15, Beth = 10, Cam = 8, Deb = 11, Eve = 4, Fran = 7, Gigi = 2, Hal = 12, Ivy = 9, Jan = 12, Kay = 6, Lexi = 8, Max = 8, Nara = 10, Omar = 5

Which booth finished last and how many minutes from the starting time did that booth finish serving its last customer?

A. Booth 1 - 42 min.
B. Booth 2 - 41 min.
C. Booth 3 - 44 min.
D. Booth 3 - 46 min.
E. Booth 2 - 45 min.

4. FSAs

Which of the following strings are accepted by this FSA?



a. 1101110 b. 1121022 c. 11021102

d. 1010 e. 1102112 f. 1011112

A. a, b, c, e
B. b, d, f
C. a, c, d
D. b, c, d
E. c, d, e, f

5. Regular Expressions

The ACSL Diner opened this year. The cook wants to order seasonings that satisfy the following regular expression. How many will he/she order?

[a-p]* e* [^s]* [d-hp-y]

relish	salt	mustard	pickle
pepper	ketchup	butter	vinegar
mayo	soy	salsa	honey

A. 4
B. 5
C. 6
D. 7
E. 8